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The two-stage RUL prediction across operation conditions using deep transfer learning and insufficient degradation data

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A R T I C L E I N F O

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A B S T R A C T

The remaining useful life (RUL) prediction provides an essential basis for improving mechanical equipment reliability. In practical application, the variant of working conditions and incomplete degradation data seriously deteriorate the performance of the prognostic models. In order to conquer this problem, a two-stage RUL prediction method is proposed for the cross-domain prognostic task with insufficient degradation data. At first, the two-level alarm mechanism is employed to detect the first predicting time (FPT) of each mechanical entity adaptively. Then, the deep separable convolutional network with the double transferable attention mechanism (DSCN-DTAM) is proposed to construct the cross-domain prognostic model. In DSCN-DTAM, multiple regularization strategies can guide the model to extract domain-invariant features, and the double transferable attention mechanism is designed to select the degradation information with high transferability. Finally, the proposed method is verified by multiple transfer prognostic tasks designed by two bearing datasets. Compared with other methods, the proposed method shows superior performance.

1. Introduction

With the rapid development of industrial Big Data and industrial Internet of Things, Prognostic and Health Management (PHM) has changed the perception of mechanical reliability engineering [1–3]. As the critical technology of PHM, the remaining useful life (RUL) prediction can estimate the time before the equipment failure and help decision-makers avoid equipment shutdown as early as possible [4]. Therefore, the RUL prediction methods are widely developed to optimize the maintenance strategy and improve the machinery reliability [5].

Recently, the two-stage framework has been extensively applied for the RUL prediction. In the two-stage framework-based RUL prediction approach, firstly, the first predicting time (FPT) identification is used to judge the start time of the mechanical degradation state [6]. Once the degradation state is detected, the RUL prediction is triggered to obtain the RUL of the mechanical equipment [7,8]. Ahmad et al. [9] employed the gradient based-method to determine FPT and used a quadratic regression model to predict bearing RUL. Pan et al. [10] identified FPT by the relative root mean square (RRMS) and predicted RUL using the extreme learning machine (ELM). In order to overcome the dependence of the data-driven approach on manual feature extraction, deep learning (DL) techniques with end-to-end learning ability have been widely applied to build the prognostic model [11,12]. Li et al. [5] determined the FPT by the 2σ criterion and proposed the multi-scale convolutional neural network (CNN) to improve the model performance for the RUL prediction. Yang et al. [13] used double CNN to simultaneously realize FPT identification and RUL prediction. Although the above studies have achieved satisfactory performance, two challenging issues still restrict the practical applications of the two-stage framework-based RUL prediction approach [14].

The first issue is that the above FPT identification methods cannot adapt to the individual differences and the random noises due to the use of the fixed alarm threshold and single identification strategy [9,13], as shown in Fig. 1. These factors may cause false alarms, thus interfering with the degradation information selection and the prognostic model construction [6]. Therefore, it is necessary to investigate a more stable FPT identification method to determine the appropriate FPT for each mechanical entity adaptively.

The second issue is that the requirement of distribution consistency limits the application of the data-driven approach in the prognostic task under multiple working conditions [15,16]. Different working conditions may cause the domain shift problem, thus inducing the performance deterioration of the prognostic model [17,18], as shown in Fig. 2

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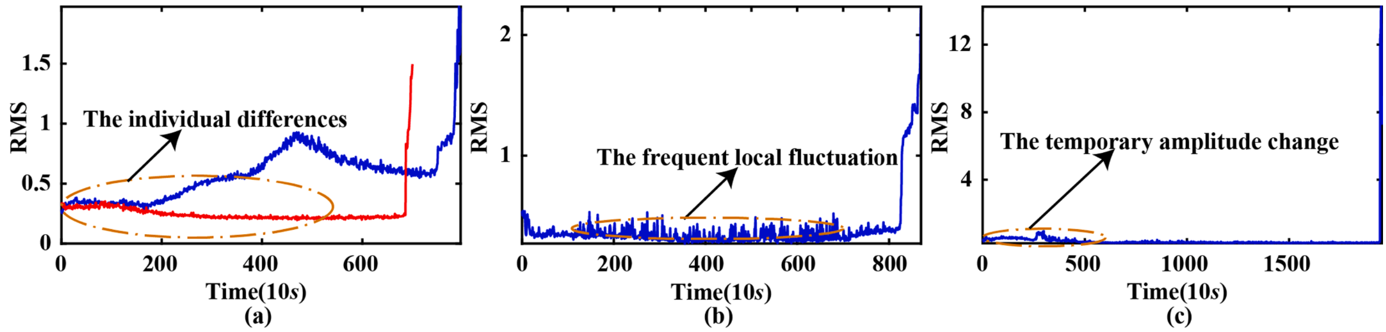

Fig. 1. The schematic representation of mechanical degradation trend affected by individual difference and random noise.

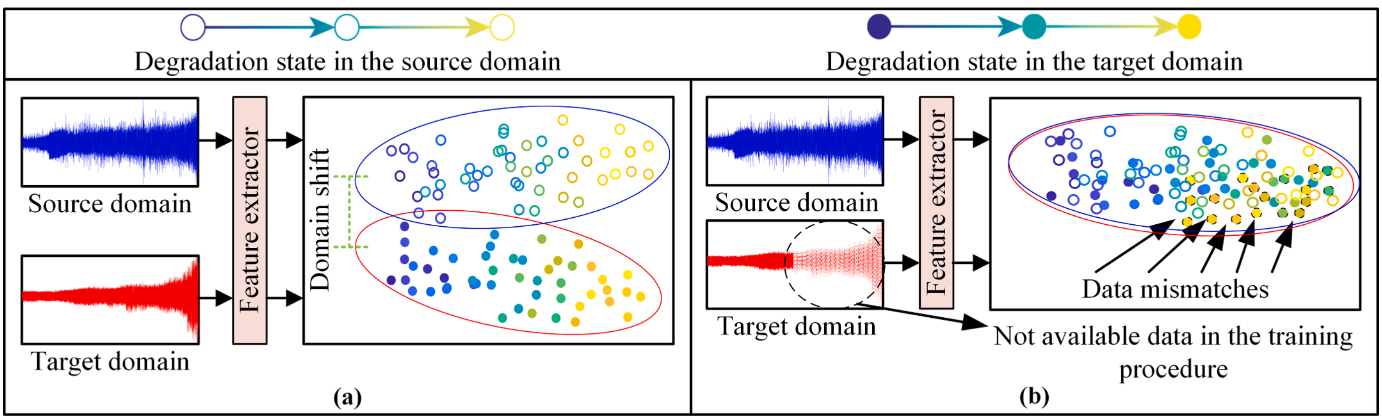

Fig. 2. The schematic representation of (a) the domain shift problem and (b) the data mismatches problem.

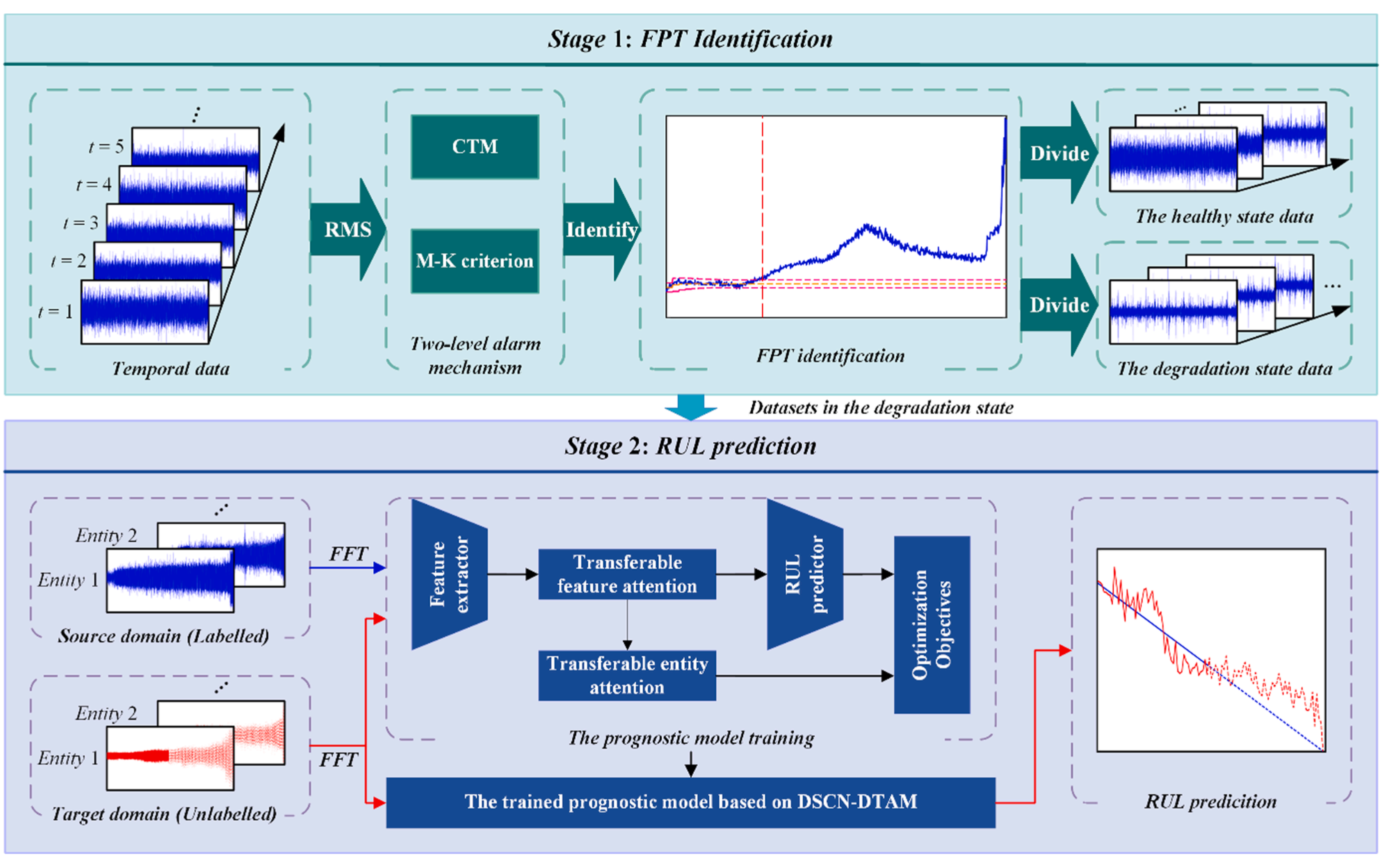

Fig. 3. The flowchart of the proposed method.

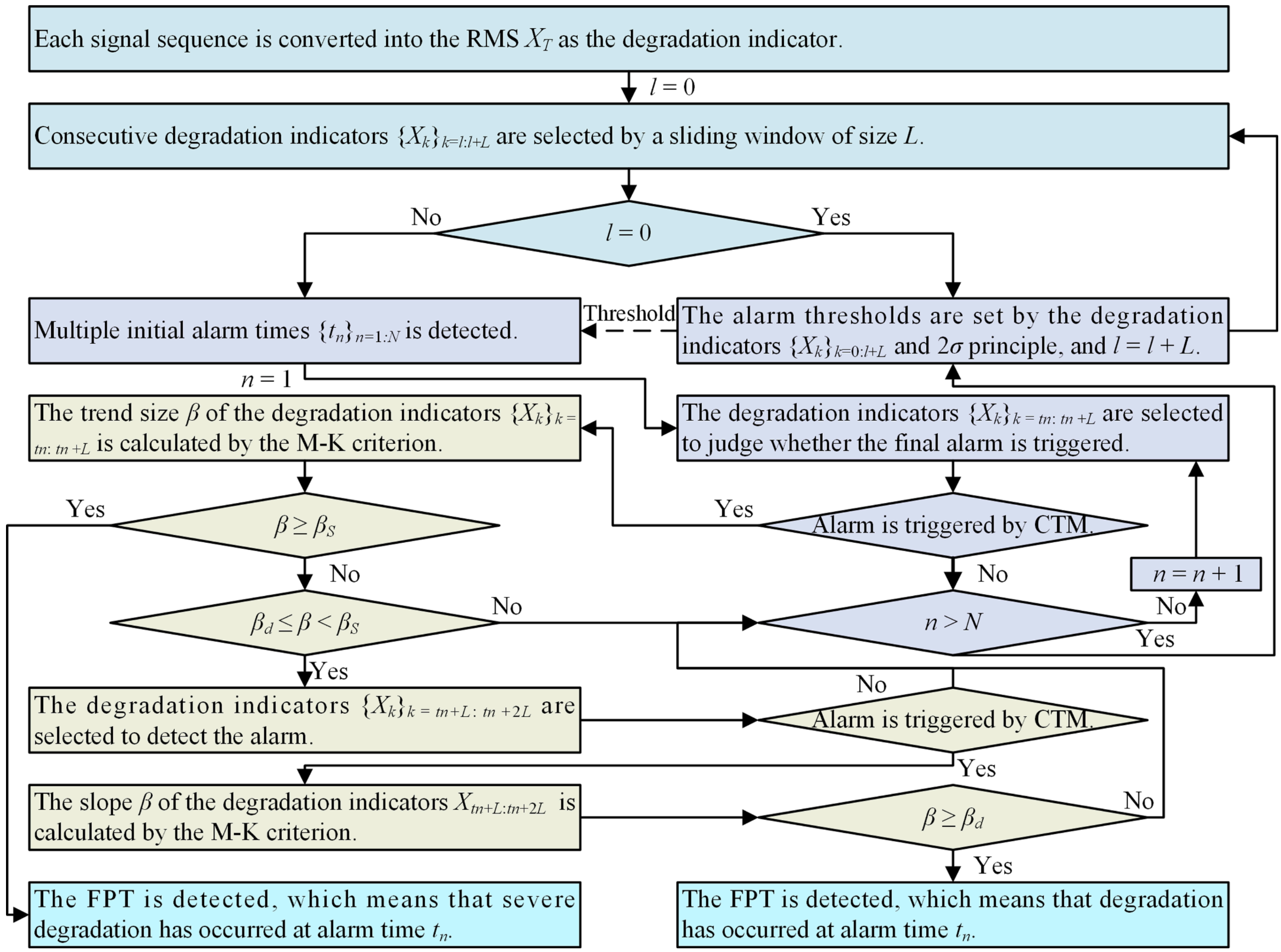

Fig. 4. The flowchart of the FPT identification method based on the two-level alarm mechanism.

(a). In order to address the domain shift problem, some scholars have carried out research focusing on the RUL prediction based on transfer learning (TL) [19,20]. For instance, Costa et al. [21] developed the prognostic method based on domain adversarial neural network (DANN) to learn the domain-invariant degradation features. Ragab et al. [22] employed the contrastive adversarial domain adaptation (CADA) method to solve the domain shift problem. Ding et al. [23] developed the deep subdomain adaptive regression network (DSARN) for bearing RUL prediction across different working conditions. From the above, the TL-based prognostic methods show great potential in cross-domain scenarios. However, some shortcomings also exist in the TL-based prognostic method: (1) The above methods cannot solve the data mismatches problem caused by only providing incomplete degradation data under the target working condition [17], as shown in Fig. 2(b). (2) Forcefully transferring the unshareable information may induce negative transfer and unsatisfactory performance in the cross-domain prognostic task [24].

In order to solve the above problems, the paper presents a novel two-stage RUL prediction approach. At first, the two-level alarm mechanism is adaptively developed to detect the FPT of each mechanical entity. In this method, false alarms can be restricted by dynamically updating the alarm threshold and simultaneously monitoring the trend and statistical characteristics of the degradation indicator. Then, the deep separable convolutional network with the double transferable attention mechanism (DSCN-DTAM) is utilized to learn the domain-invariant degradation features from complete data of a known working condition and incomplete data of an unknown working condition. In the DSCN-DTAM, multiple representation regularization strategies are designed to avoid domain shift and data mismatches. In addition, the double transferable attention mechanism is introduced to eliminate the influence of untransferable information. Finally, the extracted domain-invariant degradation features are adopted for cross-domain RUL prediction. The main contributions of this paper are summarized as follows.

(1) The two-level alarm mechanism is designed to automatically identify each mechanical entity's FPT, which has higher robustness to individual variation and random noise.
(2) A DSCN-DTAM based-prognostic method is proposed to handle domain gaps and data mismatch in the cross-domain prognostic task with incomplete degradation data.
(3) The double transferable attention mechanism is built to dynamically activate the degradation information with high transferability, thus avoiding the negative transfer problem.

The framework of this study is presented as follows. Section 2 presents the proposed two-stage RUL prediction method in detail. The proposed method is analyzed and verified through multiple transfer tasks in Section 3. Finally, Section 4 draws some conclusions.

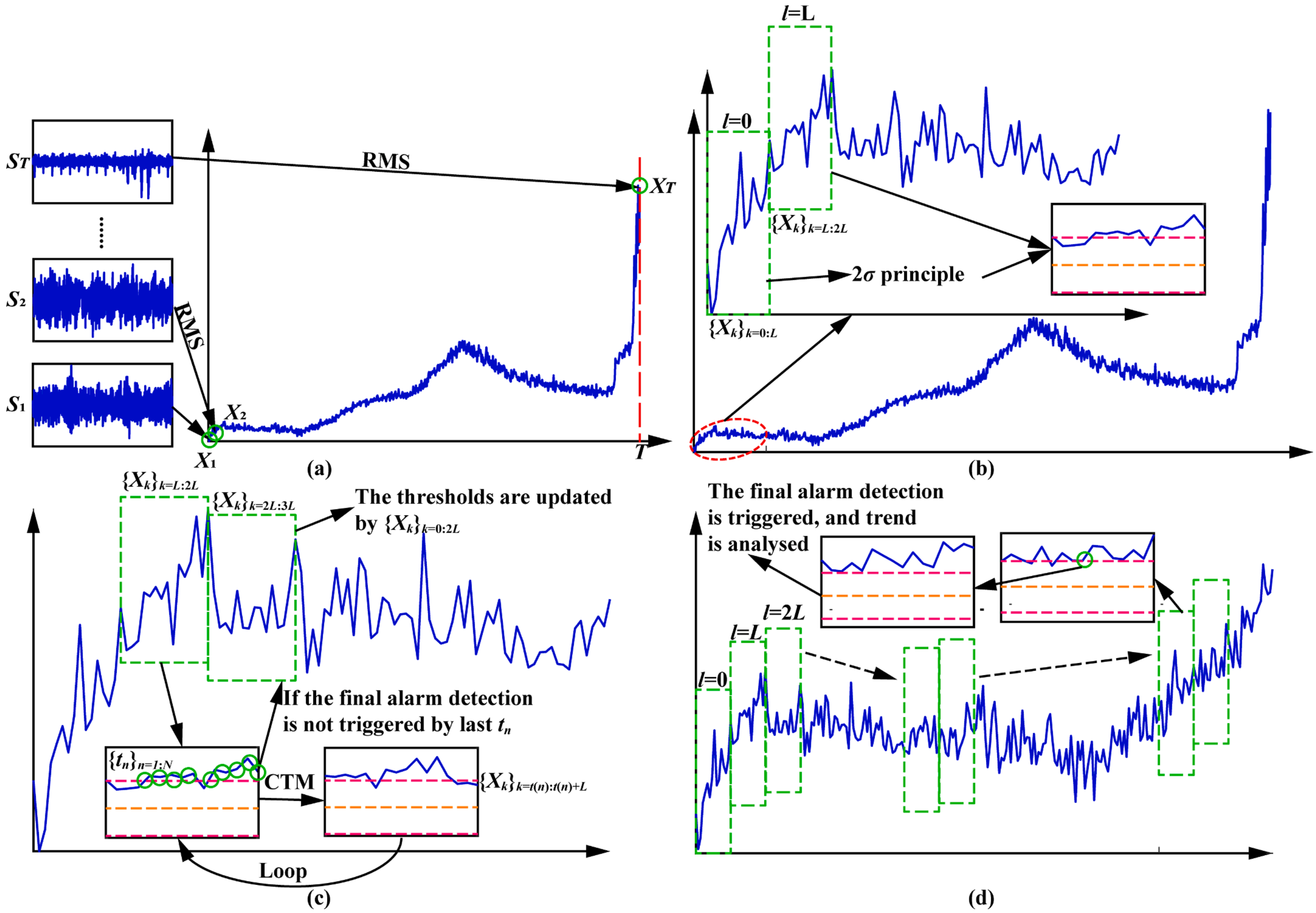

Fig. 5. The schematic illustration of the proposed FPT identification method. (a) RMS transformation, (b) The alarm thresholds setting, (c) The initial alarm detection, (d) The final alarm detection.

2. The proposed method

2.1. Problem statement

This study assumed that there are sufficient labelled full-cycle data $\{x_i^{s,j}, y_i^{s,j}\}_{i=1}^{n_{s,j}}$ of N_s mechanical entities in a known working condition (denoted as the source domain D_s), where $x_i^{s,j}$ and $y_i^{s,j}$ represent the ith data point of the jth source entity and its RUL label, respectively. $n_{s,j}$ indicates the number of data points for the jth source entity. Besides, it is assumed that only unlabeled partial life cycle data $\{x_i^{t,j}\}_{i=1}^{n_{t,j}}$ of N_t mechanical entities are collected in a new working condition (denoted as the target domain D_t), where $x_i^{t,j}$ indicates the ith data point of the jth target mechanical entity, and $n_{t,j}$ indicates the number of data points for the jth target mechanical entity. It should be pointed out that the partial life cycle data includes the data from the healthy working period and the early machine degradation period [17].

Based on the above assumptions, this paper sets two objectives. The first objective is to design an FPT identification method that can identify an appropriate FPT for each mechanical entity and resist the interference of individual variation and random noise. The second objective is to construct the cross-domain prognostic model by exploring the complete source-domain degradation data and the incomplete target-domain degradation data. In the model construction, apart from the domain shift problem, the data mismatches problem and the transferable information selection should also be addressed. Therefore, the proposed method can accurately predict the RUL of the target mechanical entity.

2.2. Methodology overview

The flowchart of the proposed method is illustrated in Fig. 3. At first, condition monitoring data are acquired from operating machinery. Then, the root mean square (RMS) is extracted from condition monitoring data to represent the degradation state of the operating machinery for FPT identification [6,7]. Next, the two-level alarm mechanism is adopted to monitor the development of RMS, thus adaptively identifying the FPT of each mechanical entity. Once the FPT is identified, the degradation state data of mechanical entities in different domains are selected to train the cross-domain RUL prediction model based on DSCN-DTAM. In the model construction, the degradation state data of the source domain are labelled as [17,20],

$$y_i^{s,j} = \frac{n_{s,j} - i}{n_{s,j} - FPT_{s,j}} \tag{1}$$

where $FPT_{s,j}$ represents the identified FPT of the jth source entity. Besides, the double transferable attention mechanism and multiple optimization objectives are designed to minimize the distribution discrepancy of the degradation state data between different domains. In this way, the domain-invariant degradation features can be extracted to improve the performance and generalization of the prognostic model on the target domain.

Table 1
The steps of the FPT identification using the two-level alarm mechanism.

Algorithm 1 The two-level alarm mechanism

Input: The vibration signal sequences $\{S_k\}_{k=1:T}$ of a mechanical entity measured at T moments; the size of the sliding window L; the trend threshold of the rapidly degradation β_S; the trend threshold of the slowly degradation β_d.
Output: FPT

Each signal sequence S_k is converted into the RMS value X_k;
Initialize $l = 0$;
Calculate the alarm threshold Th from consecutive RMS $\{X_k\}_{k=l:l+L}$ by the 2σ principle;
While FPT is none **do**
  $l = l +$ L, and select consecutive RMS $\{X_k\}_{k=l:l+L;}$
  // The initial alarm detected by CTM
  Use $|X_k|>Th$ to detect initial alarm times $\{t_n\}_{n=1:N(N\leq L)}$ from $\{X_k\}_{k=l:l+L}$;
  If $\{t_n\}$ is not none **do**
    For $n = 1:N$ **do**
      If $\forall X_k \in \{X_k\}_{k=tn:tn+L}$, $|X_k|>Th$ **do**
        // The final alarm detected by M-K criterion
        Calculate trend size β from $\{X_k\}_{k=tn:tn+L}$ by M-K criterion;
        If $\beta \geq \beta_S$ **do**
          FPT = t_n;
        Elseif $\beta_d \leq \beta < \beta_S$ **do**
          Select consecutive RMS $\{X_k\}_{k=tn+L:tn+2L}$, calculate trend size β from $\{X_k\}_{k=tn+L:tn+2L}$ by M-K criterion;
          If $\forall X_k \in \{X_k\}_{k=tn+L:tn+2L}$, $|X_k|>Th$ and $\beta \geq \beta_d$ **do**
            FPT = t_n;
          end
        end
      end
      If FPT is not none **do**
        Break;
      End
    End
  End
  Update the alarm threshold Th using RMS $\{X_k\}_{k=0:l+L}$ and the 2σ principle;
End

2.3. Stage 1: FPT identification

In general, the healthy state data provide little degradation information for RUL prediction [20]. Therefore, it is necessary to select the degradation state data through FPT identification to eliminate the interference of irrelevant information in the training process of the prognostic model. In the FPT identification, random noises caused by dust penetration or random wear during running-up may lead to false alarms, such as frequent local fluctuation and temporary amplitude change [25]. Besides, individual variation also may interfere with FPT identification because it is difficult to set a unified alarm threshold for the mechanical entities with different degradation trends [10].

This paper proposes the two-level alarm mechanism to adaptively identified FPT and overcome the interference of the above factors. The flowchart of the proposed method and the schematic illustration of key steps are shown in Fig. 4 and Fig. 5, respectively. At first, the vibration signals are collected and converted into RMS for a mechanical entity. Then, the sliding window is employed to select the consecutive RMS indicators. Noted that the data in the first window are default to the healthy state. In order to adapt to individual variation, the alarm threshold is calculated from the RMS indicators of the target machinery through the 2σ principle, without historical data. Besides, the alarm threshold can also be updated with degradation development. Next, considering that the random noises will not appear consecutively or have a long-term growth trend [5,6,9], the continuous trigger mechanism (CTM), namely the RMS indicators continuously exceed the alarm threshold, is used to judge whether to perform the initial alarm, and the Mann-Kendall (M-K) criterion is employed to analyze the trend of the RMS indicators and determine whether to perform the final alarm [26]. If two alarms are executed simultaneously, the FPT is identified. In this way, the proposed method can effectively avoid the false alarm caused by random noises. The detailed steps are listed in Table 1.

2.4. Stage 2: RUL prediction

2.4.1. The cross-domain prognostic model based on DSCN-DTAM

The cross-domain prognostic model is constructed by DSCN-DTAM, which consists of the feature extractor, the double transferable attention module, and the RUL predictor. The network structure of DSCN-DTAM is presented in Fig. 6. In order to effectively extract the high-level degradation features from the source and target domain data, the separable convolutional module is used to construct the feature extractor, which can reduce the model complexity by decoupling spatial correlation and channel correlation [27].

After extracting the high-level degradation features, the double transferable attention mechanism, which consists of the transferable feature attention and the transferable entity attention, is designed to improve the transferability of the extracted features [24,28], as shown in Fig. 7. At first, the transferable feature attention is adopted to guide the model to focus on feature regions with more transferable

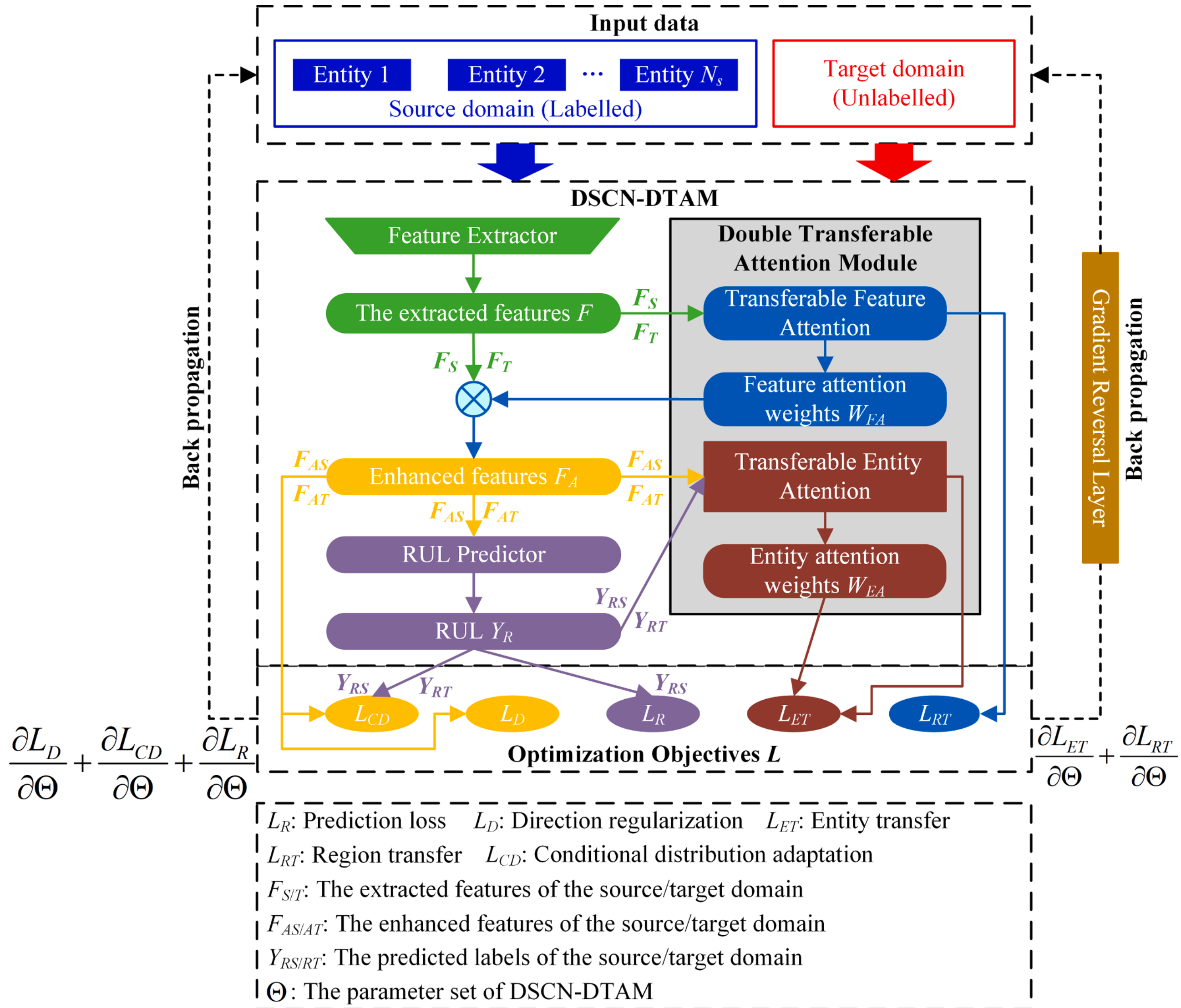

Fig. 6. The network structure of the proposed DSCN-DTAM.

degradation information [24]. It is assumed that the dimension of the extracted features F is $C \times W$, where C is the channel number and W is the feature length. The degradation features F is divided into K regions with dimension $C\times(W/K)$ along the feature direction. After that, a series of region-wise domain discriminators $\{G_{fd}^k\}_{k=1,\dots,K}$ are constructed, and each discriminator is responsible for evaluating the distribution discrepancy between the source and target domain data corresponding to the feature region k. The output of each discriminator is calculated as follows:

$$d_k = G_{fd}^k(F_k) \tag{2}$$

where F_k is the kth feature region, and d_k is the probability of the kth feature region belonging to the source domain. Next, to quantify the transferability of the kth feature region, the feature attention weight w_{FA}^k is calculated as follows:

$$w_{FA}^k = 1 - H(d_k) \tag{3}$$

where $H()$ is the entropy functional [24]. Then, the extracted features are weighted by the residual mechanism, which can avoid the negative effect of wrong feature region attention, as follows:

$$F_A^k = \left(1 + w_{FA}^k\right)\cdot F_k \tag{4}$$

where F_A is the enhanced degradation feature. In this way, the cross-domain prognostic model will focus more attention on those feature regions with greater transferability [24].

After the transferable feature attention, the enhanced degradation features are further input into the transferable entity attention. The transferable entity attention can determine which source entity information can be activated and utilized in the current transfer task [18,29]. In the transferable entity attention, N_s entity-wise domain discriminators $\{G_{ed}^n\}_{n=1,\dots,Ns}$ are employed to align the target domain data and each source entity data. In order to quantitatively evaluate the contribution of each source domain entity to the transfer task, the transferable entity attention weights are constructed by the conditional distribution discrepancy measured by the conditional embedding operator discrepancy (CEOD) [30]. In CEOD, the conditional embedding operator $\mathscr{C}_{Y|X}$ is utilized to embed the conditional distribution $P(Y|X)$ into the reproducing kernel Hilbert spaces (RKHS), and the detail is as follows [31]:

$$\begin{aligned}&\mathscr{C}_{Y|X} = \Phi(\mathrm{K} + n\eta\mathrm{I})^{-1}\Upsilon^T\\&\mathrm{K} = \Upsilon^T\Upsilon\end{aligned} \tag{5}$$

where Φ and Υ represent the operation embedding random variables X and Y into their corresponding RKHS $\mathscr{H}$ and $\mathscr{G}$. $\eta\mathrm{I}$ is the regularization term. n is the amount of data. Referring to the form of maximum mean

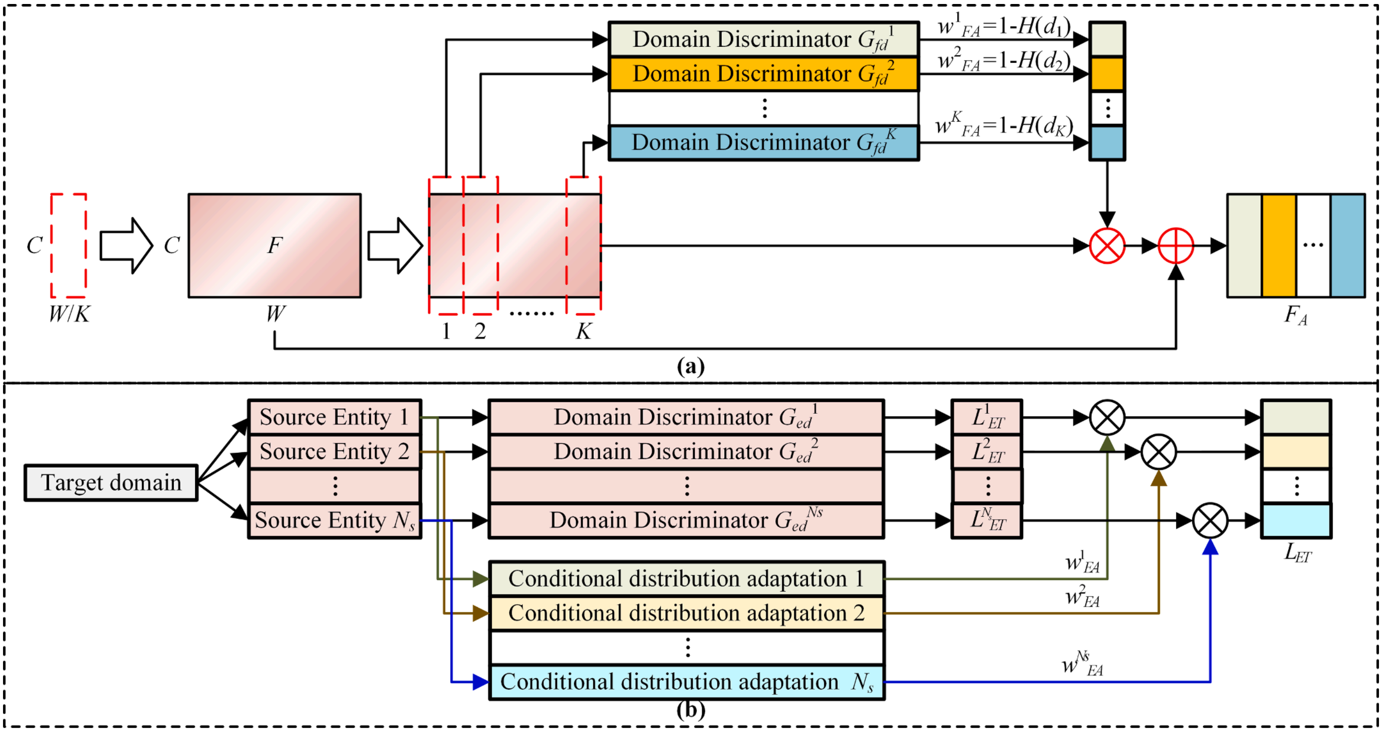

Fig. 7. The procedure of the double transferable attention mechanism. (a) The transferable feature attention, (b) the transferable entity attention.

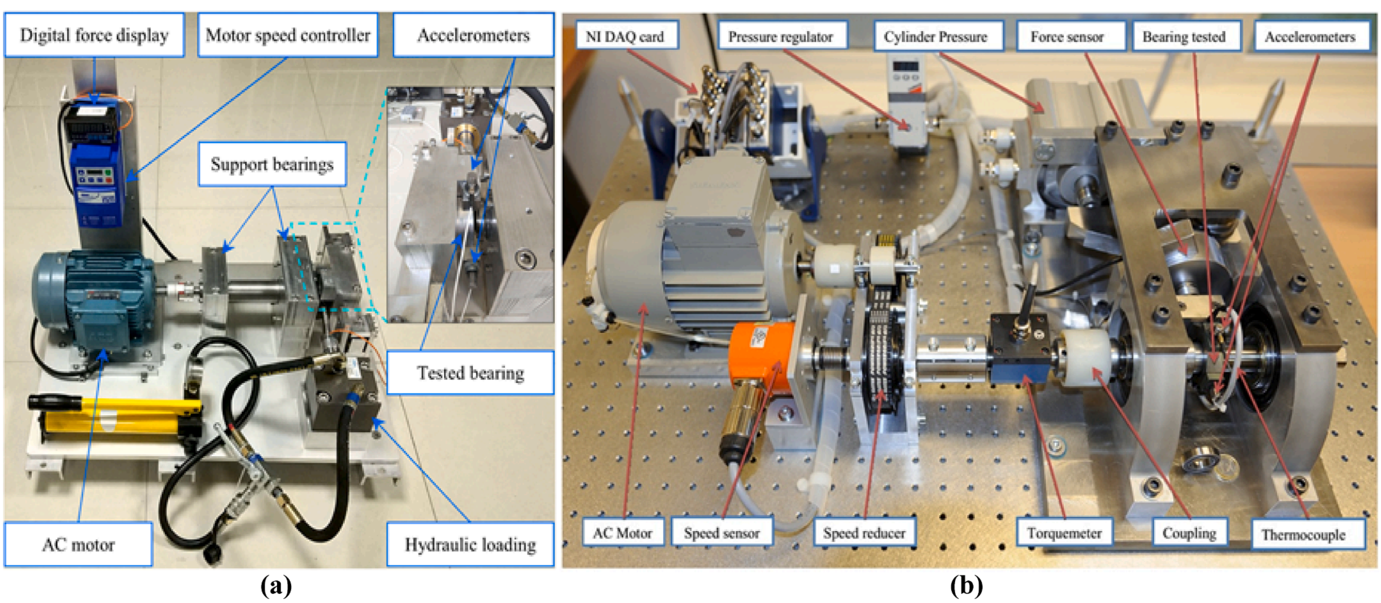

Fig. 8. The bearing platforms in (a) the XJTU-SY dataset and (b) PHM2012 dataset.

Table 2
The description of two experimental datasets.

Dataset	Name	Loading force	Rotating speed	Bearing entities
XJTU-SY	XB1	12 kN	2100 rpm	XB11, XB12, XB13, XB15
	XB2	11 kN	2250 rpm	XB21, XB22, XB23, XB24, XB25
PHM2012	PB1	4 kN	1800 rpm	PB11, PB12, PB13, PB14, PB15, PB16, PB17
	PB2	4.2 kN	1650 rpm	PB21, PB22, PB23, PB24, PB26, PB27

discrepancies (MMD), the conditional distributions discrepancy between different domains can be calculated as follow:

$$
\begin{aligned}
CEOD(D_s, D_T) &= \| \mathscr{C}_{Y_S|X_S} - \mathscr{C}_{Y_T|X_T} \|^2_{\mathscr{HS}} \\
&= \| \Phi(Y_S)(\mathrm{K}_{X_SX_S} + n_1\eta\mathrm{I})^{-1}\mathrm{Y}^T(X_S) - \Phi(Y_T)(\mathrm{K}_{X_TX_T} + n_2\eta\mathrm{I})^{-1}\mathrm{Y}^T(X_T) \|^2_{\mathscr{HS}}
\end{aligned} \tag{6}
$$

where D_s and D_T denote the conditional distribution of the source domain and the target domain, respectively. In addition, the Gaussian kernel is utilized to calculate K in this paper. Since the smaller CEOD value indicates that these two samples follow a similar conditional distribution, the attention weight is calculated as follows:

Table 3
The description of two dataset and their application in the transfer prognostic task.

Task	Transfer scenario	Training dataset in source domain	Testing dataset in target domain	Validation dataset in target domain
XB1–2	XB1→XB2	XB11, XB12, XB13, XB15	XB21, XB22, XB23, XB24	XB25
XB2–1	XB2→XB1	XB21, XB22, XB23, XB24, XB25	XB11, XB12, XB13	XB15
PB1–2	PB1→PB2	PB11, PB12, PB13, PB14, PB15, PB16, PB17	PB21, PB22, PB23, PB24, PB26	PB27
PB2–1	PB2→PB1	PB21, PB22, PB23, PB24, PB26, PB27	PB11, PB12, PB13, PB14, PB15, PB16	PB17

Table 4
Parameters used in this study.

Parameter	Value	Parameter	Value
Number of kernels in each layers	[128, 64, 32,16, 8]	Polling size	2 × 1
Size of convolution kernels	3 × 1	Number of neurons	320
Learning rate	0.0005	Batchsize	128
Epoch	600	λ_{CD}	0.01
λ_{RT}	0.1	λ_{ET}	0.001

Table 5
The description of comparison methods for FPT identification.

Comparison method	Description
The 2σ principle method [5]	This method needs preset health state data to calculate the fixed statistical threshold for FPT identification.
The gradient method with the linear rectification technique (G-LRT) [9]	This method uses the gradient of the degradation indicator to identify FPT, where the degradation indicator is adjusted by the linear rectification technique (LRT) to reduce the local fluctuations.
The gradient method with the moving average filter (G-MAF)	This method uses the gradient of the degradation indicator to identify FPT, where the degradation indicator is adjusted by the moving average filter (MAF) to reduce the local fluctuations.

$$w_{EA}^{n} = \frac{e^{-CEOD\left(D_{S}^{n},D_{T}\right)}}{\sum_{n=1}^{Ns} e^{-CEOD\left(D_{S}^{n},D_{T}\right)}} \tag{7}$$

Besides, pseudo label learning is also adopted to assist in calculating the attention weight [32]. After the calculation of the attention weight, each weight will be assigned to the corresponding entity-wise domain discriminator, thus guiding the prognostic model to know which source entity should be shared or suppressed in the optimization process. Finally, the enhanced degradation features are input into the RUL predictor built by a fully-connected layer to predict the RUL.

2.4.2. Optimization objectives and training strategy

The proposed DSCN-DTAM is optimized through the following objectives.

1) **Prediction loss:** The prediction error L_R is measured by the mean squared error (MSE), which is as follows:

$$L_R = \sum_{i=1}^{n} \left(y_{sp,i} - y_{s,i}\right)^2 \Big/ n \tag{8}$$

where $y_{sp,i}$ and $y_{s,i}$ are the predicted RUL and the real RUL corresponding to the ith training data point, respectively. Please note that the prediction loss is performed only in the source domain.

2) **Direction regularization:** Direction regularization is employed to ensure the direction consistency of the degradation features. At first, the directional vector between the degradation features F_A^k of mechanical entity k and the average initial degradation features $F_{A\text{-}init}$ of all mechanical entities is calculated, where the initial degradation feature is the extracted feature at FPT time [17]. Then the angles between all directional vector pairs are used to represent the direction regularization, which can be formulated as:

$$L_D = \sum_{k=1}^{N}\left(0.5 - \frac{0.5}{n_k}\cdot\sum_{i=1}^{n_k-1}\sum_{j=i+1}^{n_k}\frac{F_{Ai}^{k} - F_{A-init}}{\| F_{Ai}^{k} - F_{A-init} \|_2}\cdot\frac{F_{Aj}^{k} - F_{A-init}}{\| F_{Aj}^{k} - F_{A-init} \|_2}\right) \tag{9}$$

where N is the number of all entities in the source and target domains, and n_k is the training data amount in the kth mechanical entity.

3) **Conditional distribution adaptation:** In order to avoid the data mismatches problem, the conditional distribution discrepancy between domains is minimized to align the extracted features at the similar degradation level of different mechanical entities. The conditional distribution adaptation is performed by CEOD as follows:

$$L_{CD} = \frac{1}{N_s}\sum_{n=1}^{N_s} CEOD\left(D_n^S, D^T\right) \tag{10}$$

4) **Region transfer:** The binary cross-entropy is adopted to measure the domain classification loss of each region-wise domain discriminator. Besides, the minimax adversarial strategy is utilized to lead the prognostic model to learn the domain-invariant features, thus achieving effective local transfer. The loss function is written as:

$$L_{RT} = \frac{1}{K}\sum_{k=1}^{K}\frac{\sum_{i=1}^{n_k^s+n_k^t}\left[-\left[d_i^k \log \widetilde{d}_i^k + \left(1-d_i^k\right)\log\left(1-\widetilde{d}_i^k\right)\right]\right]}{n_k^s + n_k^t} \tag{11}$$

where d_i^k and $\widetilde{d}_i^k$ denote the real domain label and the classified domain label corresponding to the ith data point in the kth region, respectively.

5) **Entity transfer:** The entity attention weight is embedded into the domain classification loss of each entity-wise domain discriminator G_{ed}^n, thus performing the fined-grained transfer by emphasizing the information from the shared source entities and suppressing the information from the unshared source entities. This loss can be written as follow:

$$L_{ET} = \sum_{n=1}^{N_s} w_{EA}^n\cdot\sum_{i=1}^{N_S^n+N_T}\left[-\left[d_i \log \widetilde{d}_i + (1-d_i)\log\left(1-\widetilde{d}_i\right)\right]\right] \tag{12}$$

Through the integration of all optimization objectives, the overall learning objective of the DSCN-DTAM can be rewritten as:

$$L(\Theta) = L_R + L_D - \lambda_{RT}\cdot L_{RT} - \lambda_{ET}\cdot L_{ET} + \lambda_{CD}\cdot L_{CD} \tag{13}$$

where Θ is the parameter set of DSCN-DTAM. λ_{RT}, λ_{ET}, and λ_{CD} denote the penalty coefficients for L_{RT}, L_{ET}, and L_{CD} to tradeoff each optimization objective in the unified optimization problem. The Adam algorithm is adopted to train DSCN-DTAM [33], and the back-propagation algorithm is utilized to update the parameter set Θ as follows:

$$\Theta \leftarrow \Theta - \varepsilon\left(\frac{\partial L_R}{\partial\Theta} + \frac{\partial L_D}{\partial\Theta} - \lambda_{RT}\cdot\frac{\partial L_{RT}}{\partial\Theta} - \lambda_{ET}\cdot\frac{\partial L_{ET}}{\partial\Theta} + \lambda_{CD}\cdot\frac{\partial L_{CD}}{\partial\Theta}\right) \tag{14}$$

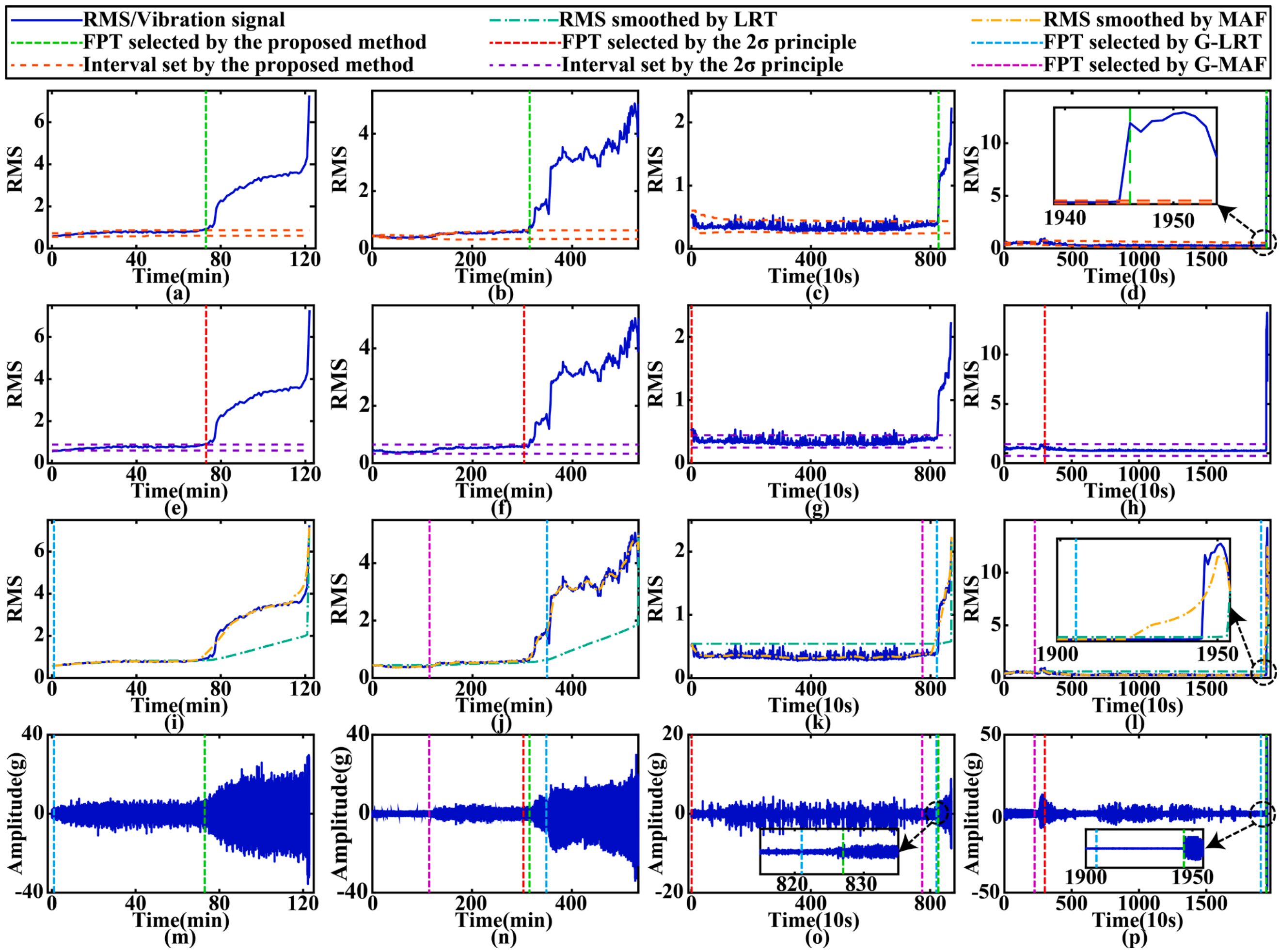

Fig. 9. The FPT results identified by different methods, including the proposed method in (a) XB11, (b) XB23, (c) PB12, and (d) PB23, the 2σ principle method in (e) XB11, (f) XB23, (g) PB12, and (h) PB23, two gradient methods in (i) XB11, (j) XB23, (k) PB12, and (l) PB23, and all FPTs in the vibration signal of (m) XB11, (n) XB23, (o) PB12, and (p) PB23.

Table 6
The comparison results of the reported FPT in publically references.

Bearing	Failure time	The proposed method	Method in [14]	Method in [25]	Method in [36]
PB11	2803	1642	1490	2118	2174
PB12	871	827	827	826	784
PB13	2375	1433	1684	1613	1842
PB14	1428	1084	1083	1082	1108
PB15	2463	2425	680	2306	1653
PB16	2448	2418	649	2035	1656
PB17	2259	2194	1026	2030	2233

where ε is the learning rate. Besides, the minimax adversarial strategy is realized by the gradient reversal layer (GRL) [34]. When the model loss L converges to an expected loss value, DSCN-DTAM can realize the RUL prediction across working conditions.

3. Experimental results and discussion

3.1. Dataset description

1) **XJTU-SY Dataset:** This dataset is shared by the Xi'an Jiaotong University and the Changxing Sumyoung Technology Company, and the experimental platform is illustrated in Fig. 8(a) [8]. This experiment tests multiple LDK UER204 bearings under various working conditions through the hydraulic loading system. The sampling frequency was 25.6 kHz, and 32,768 data points were recorded every 1 min. As given in Table 2, the horizontal vibration signals under two working conditions are used in this paper. More details are found in Ref. [8].
2) **PHM2012 Dataset:** This dataset is provided by the Franche-Comté Electronics Mechanics Thermal Science and Optics-Science and Technologies (FEMSTO-ST) Institute in the IEEE international conference PHM 2012, and the PRONOSTIA experimental platform is illustrated in Fig. 8(b) [35]. This experiment used the force actuator to accelerate the bearing degradation under different working conditions. The sampling frequency was 25.6 kHz, and 2560 data points were recorded every 10 s. The horizontal vibration signals under two working conditions are also utilized in this dataset. The detailed descriptions are presented in Table 2. Besides, compared with the XJTU-SY dataset, the individual differences of this dataset are more obvious. More details can be referred to Ref. [35].

3.2. Implementation details

This paper sets four cross-domain prognostic tasks with different source-target pairs to evaluate the proposed method. The detailed information is listed in Table 3. Note that only the health state data and the first half of the degradation state data are provided in each target domain testing dataset [17]. In each prognostic task, the vibration data are converted into RMS, and the two-level alarm mechanism is used to

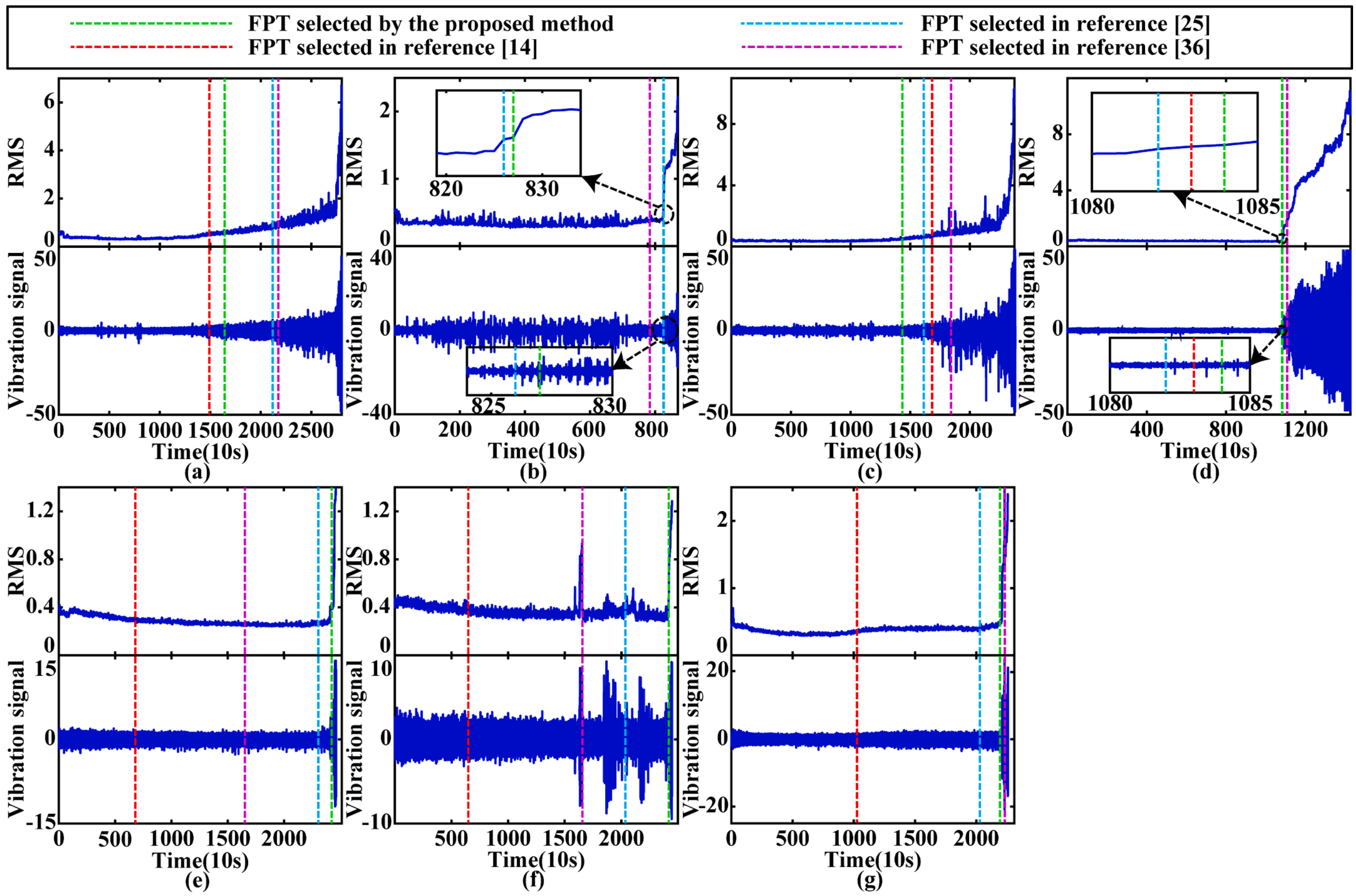

Fig. 10. FPT identification results comparison with the reported results in publically references. (a) XB11, (b) XB12, (c) XB13, (d) XB14, (e) XB15, (f) XB16, (g) XB17.

Table 7
The description of comparison methods.

Comparison method	Description
Baseline [27]	This is the conventional model without TL, which can be used to verify the transferring performance of other TL-based methods.
Direction regularization model (DR) [17]	This method is the baseline based on the direction regularization, which can be used to evaluate the benefits of the double transferable attention mechanism.
Multi-kernel MMD (MK-MMD) [18]	These methods are the domain adaptation methods, which can be used to evaluate the performance of DSCN-DTAM.
Conditional MMD (CMMD) [23]	
Global domain adversarial method (DAN) [21]	These methods are the domain adversarial methods, which can be used to evaluate the performance of DSCN-DTAM.
Conditional domain adversarial method (CDAN) [32]	

identify the FPT of each bearing, where L, β_d, and β_S are set as 15, 0.002, and 0.015, respectively. These parameters are determined by analyzing the source entity in each prognostic task. According to the identified FPT, the degradation state data of the source and target domains are selected to train the cross-domain prognostic model based on DSCN-DTAM. In this paper, DSCN-DTAM consists of one convolutional module, four separable convolutional modules, the double transferable attention module, and one fully-connected layer. Other parameters are listed in Table 4. All experiments are written in Pytorch and run five times on Windows with the Core i5–7500 CPU to reduce the effect of randomness.

3.3. Results analysis and discussion

3.3.1. Results of FPT identification

In this section, three FPT identification methods listed in Table 5 are adopted to illustrate the performance of the proposed method [5,9]. In the 2σ principle method, the preset data are selected by the FPT results identified by the proposed method. Fig. 9 shows the FPT results in four examples determined by different FPT identification methods. As shown in Fig. 9, these three methods produce false alarms in different bearing entities [7]. For example, the degradation indicators in XB11 and XB23 have a short upward trend in the early operating stage, and this trend begins to stabilize over time. This phenomenon results in the wrong identification of two gradient methods. In PB12, a significant amount of random noise leads to the failure of the 2σ principle method. Similar problems also cause poor performance of these methods in PB23. Therefore, compared with these three methods, the proposed method has a better anti-interference ability to avoid the impacts of anomaly conditions.

Besides, three FPT identification methods reported in references [14, 25,36] are employed to evaluate further the proposed method, in which references [14,36] adopt the classification approach. The publicly available FPT results of the PHM2012 dataset are summarized in Table 6, and Fig. 10 is used for a more intuitive comparison of results. As seen by the results, the FPT results identified by the proposed method are closer to the time when the bearing enters the degradation state. Therefore, the proposed FPT identification method can accurately capture the trend change of the degradation indicators, thus effectively determining the FPT.

Table 8
The comparison results of the target domain testing dataset in the XJTU-SY dataset.

Task	Bearing	Metric	Proposed	Baseline	DR	MK-MMD	CMMD	DAN	CDAN
XB	XB11	MAE	**18.3 ± 1.6**	22.2 ± 11.0	21.5 ± 6.6	18.4 ± 1.9	21.7 ± 1.3	20.9 ± 0.6	21.8 ± 2.6
2–1		RMSE	**21.6 ± 2.0**	27.0 ± 13.0	25.4 ± 7.0	22.5 ± 2.9	25.2 ± 1.6	24.4 ± 0.9	25.3 ± 3.5
	XB12	MAE	**14.8 ± 1.8**	22.0 ± 5.7	19.4 ± 3.6	18.3 ± 2.1	17.8 ± 2.6	21.5 ± 0.8	22.7 ± 2.2
		RMSE	**16.1 ± 1.2**	29.1 ± 7.8	22.8 ± 4.0	23.0 ± 2.4	20.4 ± 2.7	24.3 ± 0.7	26.2 ± 3.3
	XB13	MAE	**12.3 ± 0.9**	23.3 ± 7.1	14.0 ± 2.8	24.7 ± 3.1	17.3 ± 4.2	18.1 ± 1.9	20.3 ± 2.5
		RMSE	**15.1 ± 1.2**	27.6 ± 8.3	17.0 ± 3.8	27.8 ± 3.0	20.3 ± 5.2	21.8 ± 1.6	26.4 ± 7.2
XB	XB21	MAE	**10.5 ± 2.1**	19.2 ± 3.0	16.6 ± 4.2	15.7 ± 2.3	20.9 ± 2.7	17.7 ± 0.9	20.3 ± 1.5
1–2		RMSE	**13.0 ± 2.9**	25.1 ± 6.7	20.1 ± 4.6	19.4 ± 2.5	20.1 ± 7.2	24.6 ± 5.4	23.8 ± 1.4
	XB22	MAE	**11.4 ± 3.6**	22.1 ± 3.8	16.3 ± 4.3	22.0 ± 1.1	18.7 ± 2.1	20.2 ± 1.5	21.3 ± 0.8
		RMSE	**14.3 ± 4.1**	26.0 ± 4.7	19.9 ± 5.1	25.3 ± 1.1	26.0 ± 9.0	27.2 ± 5.3	24.6 ± 0.8
	XB23	MAE	**15.2 ± 1.0**	25.3 ± 1.5	18.1 ± 2.4	19.7 ± 1.7	24.0 ± 1.1	23.1 ± 0.3	22.7 ± 1.1
		RMSE	**18.2 ± 1.5**	31.4 ± 2.5	22.0 ± 3.3	22.9 ± 1.7	23.0 ± 8.8	23.1 ± 6.5	26.8 ± 1.0
	XB24	MAE	20.6 ± 2.2	30.7 ± 3.6	23.3 ± 4.7	22.2 ± 5.2	23.3 ± 0.7	**19.9 ± 0.5**	21.3 ± 1.1
		RMSE	23.4 ± 3.1	36.3 ± 3.5	27.2 ± 4.5	26.6 ± 6.0	**23.3 ± 7.5**	24.8 ± 2.8	24.4 ± 0.9

Table 9
The comparison results of the target domain testing dataset in the PHM2012 dataset.

Task	Bearing	Metric	Proposed	Baseline	DR	MK-MMD	CMMD	DAN	CDAN
PB	PB11	MAE	**22.9 ± 3.3**	38.2 ± 3.3	30.5 ± 4.8	35.1 ± 3.4	26.3 ± 1.3	28.0 ± 3.7	23.8 ± 2.4
2–1		RMSE	**27.2 ± 3.9**	45.6 ± 3.2	36.8 ± 5.6	42.9 ± 4.5	31.1 ± 2.1	32.5 ± 4.6	28.1 ± 2.9
	PB12	MAE	**15.5 ± 2.8**	21.4 ± 3.8	21.3 ± 3.7	24.1 ± 3.9	26.0 ± 1.7	26.4 ± 4.3	21.9 ± 5.5
		RMSE	**19.9 ± 3.3**	26.0 ± 4.4	24.6 ± 3.7	27.8 ± 4.9	30.0 ± 2.2	30.7 ± 5.3	26.8 ± 7.3
	PB13	MAE	**15.4 ± 2.6**	28.6 ± 6.1	18.2 ± 3.4	26.2 ± 2.9	23.9 ± 1.3	25.8 ± 4.4	22.9 ± 2.1
		RMSE	**18.9 ± 3.1**	37.1 ± 8.8	22.6 ± 3.9	32.3 ± 2.9	29.4 ± 2.1	32.3 ± 7.4	27.6 ± 2.6
	PB14	MAE	38.5 ± 1.6	43.0 ± 2.8	39.0 ± 3.1	39.7 ± 2.2	38.8 ± 3.0	**35.9 ± 5.2**	39.4 ± 8.6
		RMSE	45.6 ± 1.7	53.3 ± 5.8	46.2 ± 2.7	47.3 ± 1.8	51.0 ± 5.6	**43.6 ± 8.3**	53.0 ± 6.1
	PB15	MAE	**15.2 ± 4.0**	19.4 ± 1.6	16.6 ± 2.9	21.6 ± 2.7	25.0 ± 1.9	24.1 ± 6.5	25.5 ± 4.4
		RMSE	**18.2 ± 4.3**	23.4 ± 2.1	20.0 ± 3.5	25.9 ± 4.4	29.3 ± 2.6	28.3 ± 6.1	29.6 ± 4.8
	PB16	MAE	**16.2 ± 2.9**	21.5 ± 1.8	18.8 ± 1.3	22.7 ± 2.4	26.6 ± 3.4	26.1 ± 5.2	23.4 ± 2.5
		RMSE	**19.7 ± 3.8**	26.7 ± 2.0	22.6 ± 1.7	26.3 ± 3.0	31.3 ± 4.1	31.6 ± 6.4	27.8 ± 3.0
PB	PB21	MAE	**14.1 ± 1.3**	16.4 ± 2.2	14.9 ± 4.6	21.1 ± 2.3	25.2 ± 1.8	20.5 ± 2.6	26.2 ± 2.0
1–2		RMSE	**17.5 ± 1.0**	20.4 ± 2.6	18.6 ± 4.9	24.7 ± 2.7	28.8 ± 2.1	25.3 ± 4.0	34.0 ± 4.1
	PB22	MAE	24.7 ± 0.9	47.5 ± 4.3	30.2 ± 9.6	**23.9 ± 1.1**	25.4 ± 0.8	28.6 ± 6.1	25.6 ± 1.7
		RMSE	31.4 ± 1.5	53.1 ± 3.8	35.6 ± 9.3	**29.3 ± 1.2**	31.0 ± 1.7	34.2 ± 7.3	29.7 ± 1.3
	PB23	MAE	**20.7 ± 2.3**	34.1 ± 4.9	33.1 ± 4.7	32.8 ± 4.9	27.5 ± 4.5	36.6 ± 5.3	38.2 ± 6.6
		RMSE	**31.4 ± 2.1**	40.7 ± 6.0	37.8 ± 5.2	37.6 ± 4.9	31.9 ± 3.9	43.2 ± 6.1	44.4 ± 7.5
	PB24	MAE	**22.7 ± 2.9**	26.0 ± 0.3	24.7 ± 3.3	27.7 ± 1.8	32.7 ± 2.3	27.2 ± 3.4	25.3 ± 0.4
		RMSE	**24.4 ± 2.5**	30.3 ± 2.3	28.3 ± 3.1	32.8 ± 2.6	39.1 ± 3.9	31.2 ± 3.1	31.0 ± 1.9
	PB26	MAE	**22.6 ± 2.0**	26.5 ± 1.9	27.6 ± 3.3	24.7 ± 1.3	28.5 ± 9.3	23.6 ± 1.8	32.0 ± 2.3
		RMSE	**27.3 ± 1.2**	31.0 ± 3.1	33.9 ± 4.4	27.6 ± 1.3	34.0 ± 9.9	27.8 ± 2.1	38.0 ± 3.8

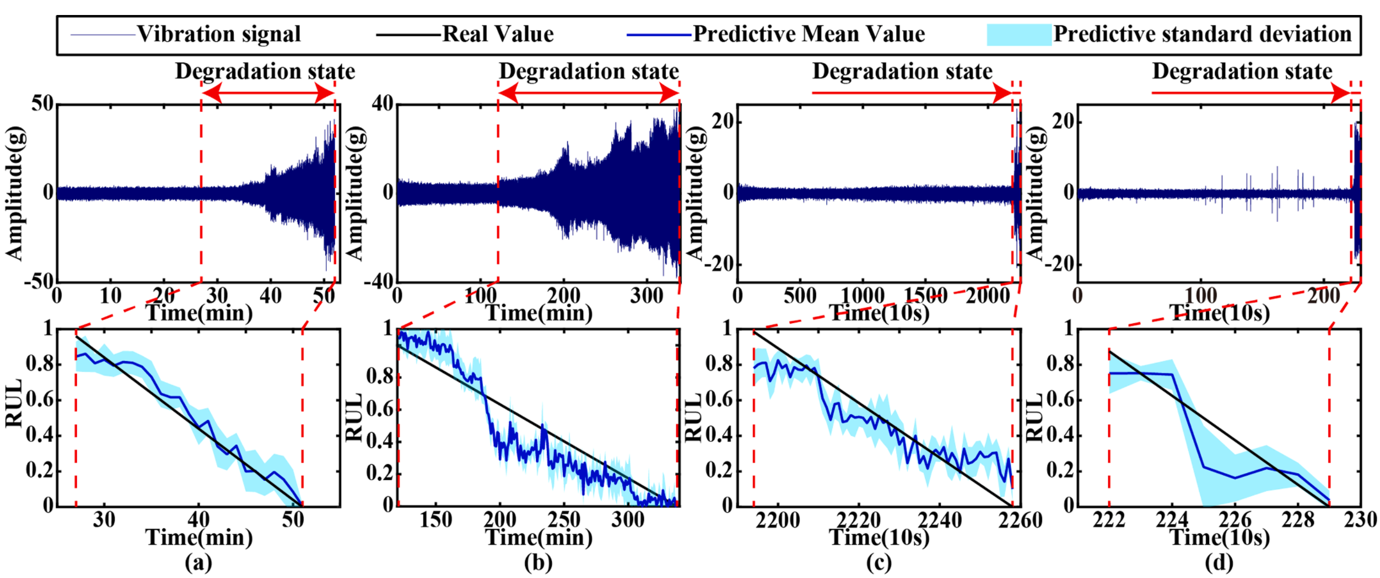

Fig. 11. RUL prediction results of the target domain validation dataset obtained by the proposed method in different tasks. (a) XB15 in task XB2–1. (b) XB25 in task XB1–2. (c) PB17 in task PB2–1. (d) PB27 in task XB1–2.

Table 10
The comparison results of the target domain validation dataset in both the XJTU-SY dataset and the PHM2012 dataset.

Task	Bearing	Metric	Proposed	Baseline	DR	MK-MMD	CMMD	DAN	CDAN
XB2–1	XB15	MAE	**7.8 ± 1.7**	16.8 ± 3.1	13.9 ± 2.8	11.4 ± 4.2	13.8 ± 3.0	12.2 ± 1.1	11.9 ± 1.7
		RMSE	**9.5 ± 2.0**	20.5 ± 3.9	16.9 ± 3.8	13.9 ± 4.4	16.6 ± 4.3	17.0 ± 1.6	15.9 ± 2.2
XB1–2	XB25	MAE	**13.5 ± 3.3**	21.1 ± 4.6	16.9 ± 3.4	18.3 ± 2.9	18.1 ± 1.5	18.5 ± 1.7	20.3 ± 2.6
		RMSE	**16.5 ± 3.3**	31.0 ± 9.6	20.4 ± 4.6	23.3 ± 3.8	21.6 ± 2.0	22.0 ± 0.5	24.0 ± 2.8
PB2–1	PB17	MAE	**12.4 ± 1.8**	25.7 ± 3.8	18.2 ± 1.8	24.3 ± 4.7	20.6 ± 2.2	20.8 ± 6.7	22.3 ± 4.2
		RMSE	**15.5 ± 2.1**	30.8 ± 3.9	22.5 ± 2.9	30.9 ± 5.8	24.5 ± 2.2	25.1 ± 7.7	27.1 ± 5.1
PB1–2	PB27	MAE	**19.0 ± 1.8**	23.1 ± 1.9	19.4 ± 2.3	27.0 ± 3.4	20.7 ± 1.8	31.2 ± 2.9	28.0 ± 6.2
		RMSE	**23.7 ± 1.9**	27.6 ± 1.5	24.7 ± 1.9	33.1 ± 5.4	25.8 ± 2.2	35.7 ± 2.6	32.5 ± 6.6

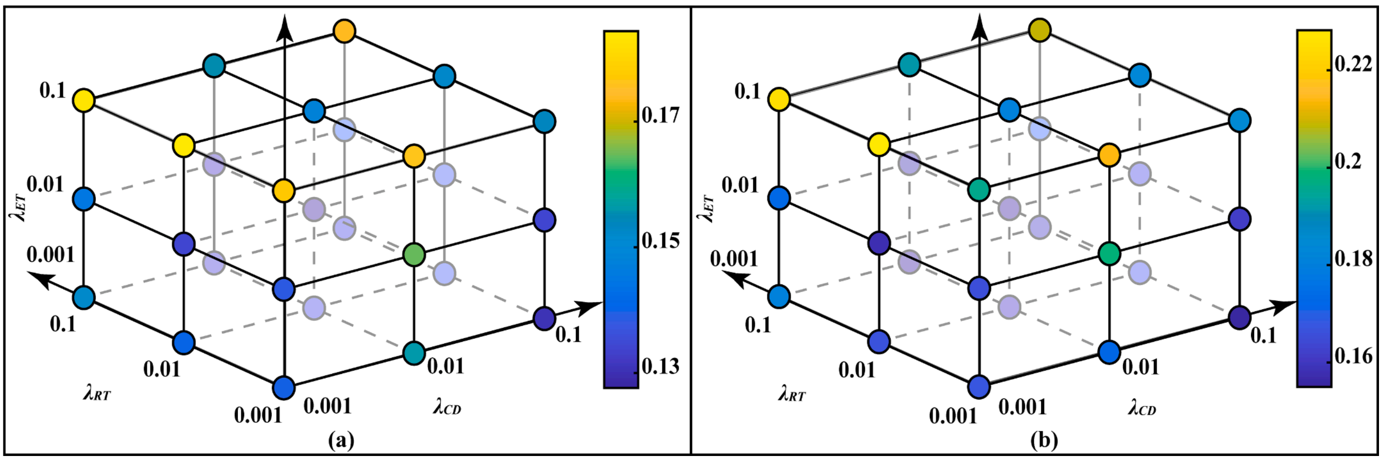

Fig. 12. The comparison results of different weight coefficients on the performance of the proposed method in task XB2–1. (a) MAE, (b) RMSE.

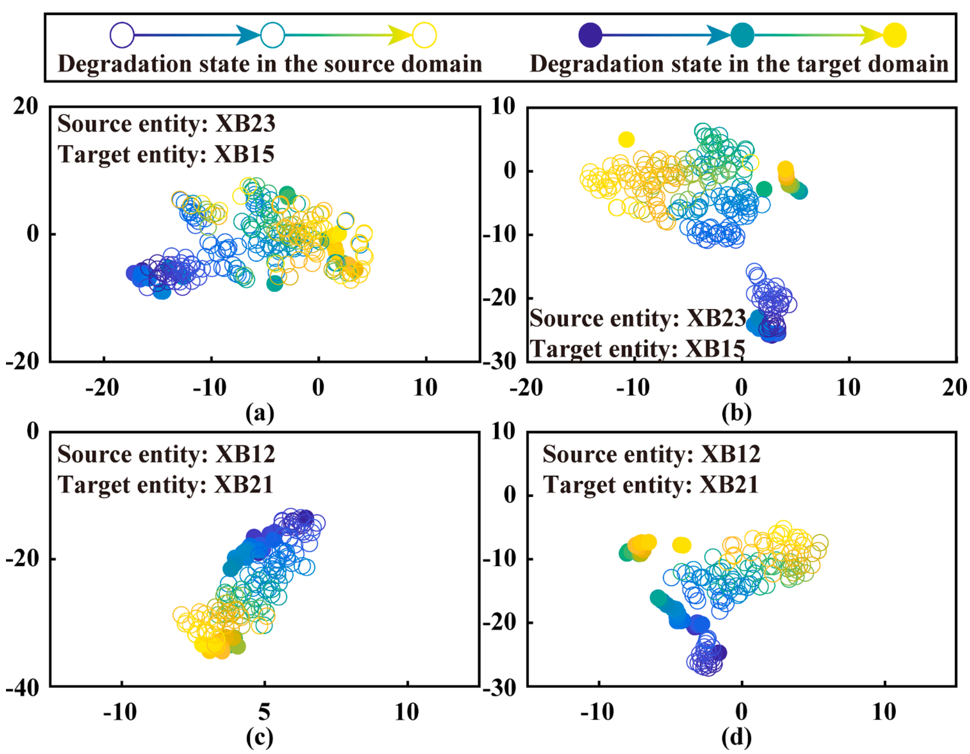

Fig. 13. Visualization results of the degradation features extracted by DSCN-DTAM and the baseline apporach. (a) DSCN-DTAM in task XB2–1, (b) The baseline approch in task XB2–1, (c) DSCN-DTAM in task XB1–2, (d) The baseline approach in task XB1–2.

3.3.2. Results of RUL prediction

3.3.2.1. Results analysis and performance comparison. In this study, six prognostic methods based on different transfer strategies are implemented on the cross-domain prognostic task for comparison study. The details of these six methods are listed in Table 7. All prognostic methods share the same network, hyperparameters and experimental setting with DSCN-DTAM. Besides, the mean absolute error (MAE) and the root mean square error (RMSE) are used to quantize the model performance.

At first, since only partial unlabeled target domain datasets are utilized in model training, both the full-cycle degradation data and their labels of the target domain testing dataset are given to evaluate the performance of different prognostic models. The numerical results of DSCN-DTAM and six state-of-the-art prognostic methods in the target domain testing dataset of XJTU-SY and PHM2012 are given in Table 8 and Table 9, respectively. In the XJTU-SY dataset, it is clear that the evaluation metrics of the prognostic method based on DTL are smaller than the baseline approach, which indicates the effectiveness of DTL in the prognostic task across working conditions. However, the negative transfer problem occurs in the domain adaptation method and the domain adversarial method for the prognostic tasks in the PHM2012 dataset. This phenomenon is because the PHM2012 dataset not only has the distribution differences caused by different working conditions but also has more significant variations in the degradation behaviors and the failure states between different bearing entities, which brings difficulties to the cross-domain transfer prognostic task with the incomplete degradation data in the target domain. Nevertheless, the proposed prognostic method still shows better prognostic performance than other state-of-the-art methods, which indicates that DSCN-DTAM can perform more fined-gained transfer, thus effectively avoiding the mismatches problem and the negative transfer.

Secondly, four validation datasets of different transfer tasks not utilized in the training process are employed to further evaluate the performance of different prognostic methods. According to the identified FPT of each bearing, the degradation state data on each target domain testing dataset are input to the trained cross-domain prognostic model, thus evaluating RUL. The prognostic results of DSCN-DTAM on the target domain testing datasets are shown in Fig. 11. As is evident from Fig. 11, the overall variation trend of the predicted results is similar to the real RUL labels with increasing operation time. Besides, Table 10 lists all numerical results of different prognostic methods. Compared with the baseline approach, the prognostic methods based on DTL can improve the prognostic performance. However, some methods still present the negative transfer problem in the task PB1–2. From the overall perspective, domain adversarial methods' performance worsens than domain adaptation methods, and the conditional distribution alignment methods perform slightly better than the marginal distribution alignment methods. Compared with these four methods, the DR. method achieves smaller prognostic errors, proving that DR. can effectively improve the performance of the cross-domain prognostic model. However, the proposed method achieves the lowest prediction errors over the six above-mentioned prognostic methods in different cross-domain prognostic tasks. Therefore, these results show that the proposed RUL prediction method can effectively bridge the cross-domain gap and handle feature distribution shift across domains in the case of incomplete target domain datasets.

3.3.2.2. Performance evaluation and feature visualization. Firstly, taking the task XB2–1 as an example, the impact of different penalty items on the DSCN-DTAM's performance is evaluated by analyzing the performance metrics of DSCN-DTAM with different weight combinations on target domain testing datasets and validation datasets. The average value is shown in Fig. 12, in which the circle's color represents the value of the performance metric. From the plot, except for $L_{ET} = 0.1$, the performance metric of DSCN-DTAM with different weight combinations does not change significantly in task XB2–1, which indicates that the prognostic method presented in this paper has relatively stable performance. Besides, compared with the average performance of six prognostic methods in Table 8 and Table 10, the performance variation range of the prognostic model with these weight combinations is acceptable. Therefore, the weight setting used in this paper is reasonable.

Furthermore, the degradation features learned by DSCN-DTAM and the baseline approach are visualized by the t-distributed stochastic neighbor embedding (t-SNE) method to validate the transfer learning effects. Taking tasks XB2–1 and XB1–2 as examples, the visualization results are presented in Fig. 13, in which the circle's color represents the degradation degree. From the plot, the degradation features extracted by the baseline approach are separated in the hidden representation space, demonstrating the distribution discrepancy's effect on the baseline approach. Compared with the baseline approach, DSCN-DTAM can effectively shorten the distribution distance in the feature sub-space, thus making the cross-domain high-level degradation features cluster in the same area. Besides, the cross-domain degradation features in the similar degradation level are also well-aligned, which indicates that DSCN-DTAM can perform more precise domain alignment. Therefore, DSCN-DTAM can effectively solve the distribution shift problem of the cross-domain RUL prediction.

4. Conclusion

This paper proposes a two-stage RUL prediction method for the cross-domain RUL prediction with incomplete degradation data of the target domain. At first, the two-level alarm mechanism is employed to adaptively determine each mechanical entity's FPT. Then, DSCN-DTAM is proposed to construct the cross-domain prognostic model. In DSCN-DTAM, the double transferable attention mechanism is adopted to dynamically activate the feature regions and source entities with high transferability, thereby avoiding the negative transfer caused by the untransferable information. Besides, multiple regularization strategies are used to guide the prognostic model to extract the domain-invariant degradation features from the incomplete cross-domain degradation data so that the model can eliminate the impact of domain shift and data mismatches. Finally, four transfer prognostic tasks constructed from two bearing datasets are adopted for performance verification. By the comparison study, the proposed two-stage prognostic method not only has a more robust performance to random noises and individual variation for FPT identification, but also has excellent and stable cross-domain prognostic performance. Therefore, the proposed method can effectively improve the prognostic performance in practical applications.

The possible drawback of this study is that the proposed DSCN-DTAM has a higher computational cost due to more model parameters. Therefore, future studies will further explore model compression technology to solve this limitation.

CRediT authorship contribution statement

Han Cheng: Methodology, Software, Writing – original draft, Visualization, Writing – review & editing. **Xianguang Kong:** Resources, Supervision, Funding acquisition. **Qibin Wang:** Conceptualization, Writing – review & editing, Funding acquisition. **Hongbo Ma:** Funding acquisition. **Shengkang Yang:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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